Phase I Swindon EV

Flood Risk Assessment

Swindon EDA Consortium

Brookbanks

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Contents

	Executive summary	
	Scoping Note Objective Summary	
1	Introduction	1
2	Background Information	1
3	Flooding Risk	7
4	Storm Drainage	24
5	Foul Drainage	37
6	Summary	40
7	Limitations	41

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Appendix 1: Drawings and Calculations

Site Survey	Drawing 1335/SV/01
Existing Drainage Regime	Drawing 1335/DR/01
Primary Drainage Feature Layout	Drawing 1335/DR/02
Secondary Drainage Feature Layout	Drawing 1335/DR/03
Tertiary Drainage Feature Layout	Drawing 1335/DR/04

Bureau Veritas Flood Zone Plan

IoH124 Output Summary

WinDES Detention Routing Calculations

Surface Water Flood Mapping

Surface Water Hazard Mapping

Appendix 2: Additional Information

Flood Risk Assessment Scoping Note

EA Scoping Note Approval Responses

Parish Council Flood Photo Matrix

Honda UK Pond Design Report

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Executive Summary

The proposed development lies to the east of Swindon. The A420 and the Great Western Rail Line bound the site to the south. To the west of the site, Thornhill Lane forms the site boundary, with the extents of South Marston, Nightingale Lane and the accompanying farmland bounding the site to the north. Agricultural land forms the eastern boundary of the site. The majority of the land is thought to be historically undeveloped, however, small developments including a combination of dwellings, hotels and public houses are spread around the site, with link roads included throughout.

The proposals aim to provide benefits to the wider Swindon area by developing a sustainable urban extension that will provide a significant contribution to the requirements of both the Swindon LDF and LP documents.

The development flood risk and drainage proposals have been scoped and agreed with the Environment Agency (EA) and various regulatory consultees before production of this FRA in a document that is contained in Appendix 2 of this report along with the respective responses.

The majority of the proposed development is shown to lie within Flood Zone 1 with a strip of land running from the north to the east lying within Flood Zones 2 and 3 of the South Marston Brook. The eastern boundary is also reported to lie within Flood Zones 2 and 3 of the River Cole.

It is proposed the development will employ a multi-tier SuDS water management train, providing source control measures implemented as close as possible to source, together with more strategic local, secondary, and regional, tertiary, treatment systems.

Implementing a tiered water management train will allow the development generally to provide three stages of water treatment, thereby minimising flood risk in potential design exceedance situations and reducing pollutant loads within the stormwater run-off by passive treatment, thus ensuring the development proposals have no adverse affects on the wider hydrology.

The systems have been designed to control the stormwater run-off from the site to circa 65.5% below the baseline greenfield rate as calculated using the IoH124 methodology. The site will therefore deliver a valuable reduction in peak storm discharges to the wider catchment across the area, which is know to suffer a number of flooding problems,

All design measures developed for the scheme have been appraised for the potential effects of climate change over the lifetime of the development and the output calculations show the proposals to accommodate potential increases in rainfall without failure.

Foul drainage on site will be discharged to the existing Thames Water Network via one of a number of potential options that will have capacity to accommodate flows from the proposed development following completion of a detailed modelling study by Thames water.

The sewer network across the site will be designed in accordance with Sewers for Adoption and adopted under Section 104 of the Water Industry Act. The above ground storm water management systems will be adopted by Swindon Borough Council or another suitable body. The ongoing maintenance of the site drainage system will therefore be assured.

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Scoping Note Objective Summary

Prior to undertaking this FRA, a scoping document was prepared and submitted to a working group of regulatory bodies, the aim being to agree the scope and fundamental outputs for this report and associated proposals. This document has since been approved by the Environment Agency and used to inform this FRA. The scoping note can be found in Appendix 2, along with the EA correspondence. Within the scoping note, a series of key objectives for this FRA were identified. Figure i below provides a summary of these objectives along with the location within this FRA compliance with these is highlighted.

Number	Objective	Location within FRA
1	Development at Swindon EDA is to have regard to the Swindon Borough Council Strategic Flood Risk Assessment with proposals providing specific compliance with the SFRA policies.	Pages 7, 8 & 22
2	The FRA will undertake a comprehensive assessment of the South Marston catchment mechanisms and functions in order to understand the types of flood risk associated with this site	FRA Section 3
3	The Phase I Swindon EV development proposals will utilise the Environment Agency Flood Map for the 1 in 100 and 1 in 1,000 year flood zone extents as the baseline for the site with regard to fluvial flood risk.	Pages 9 & 10
4	The FRA is to have regard to the localised surface water flooding incidents identified by the Parish Council and will introduce on site mitigation to address surface water flooding, and off site mitigation where possible.	Pages 10 – 16; Photo Matrix in Appendix 2
5	The FRA is to appraise flood risk from all potential forms of flooding including fluvial, overland flow, ground water, sewers and artificial water bodies using information from available sources.	FRA Section 3
6	A Sequential Test and Exception Test will be completed to demonstrate appropriateness of the proposed development in the context of NPPF Guidance	Pages 18 - 21
7	The storm drainage system at the site is to be designed in accordance with NPPF, SFRA, Sewers for Adoption and published CIRIA guidance et al	FRA Section 4
8	Development proposals will use the IoH124 figures calculated in Figure 4a. The drainage system will be designed in accordance with the principles set out in CIRIA 697. Pre development Greenfield run-off rates and a clear percentage reduction will be agreed with the Environment Agency. The proposed development will therefore aim to deliver a valuable reduction in surface water run-off to benefit the wider catchment	FRA Section 4
9	SuDS systems are to be designed to accommodate the potential effects of climate change in accordance with NPPF	FRA Section 4
10	Where possible, SuDS systems are to be unlined thereby providing limited infiltration to ground although as a good practice measure they will be designed assuming no infiltration	Pages 21 & 24
11	SuDS systems are to be designed to minimise future maintenance and, prior to any reserved matters application, an appropriate adopting authority will be appointed in accordance with a long term maintenance plan	Pages 35 & 36
12	The SuDS system will be designed to promote biodiversity and improve water quality	Pages 34 & 35
13	The FRA is to investigate capacity in the existing foul drainage network and promote a system that accords with Sewers for Adoption and associated standards to ensure adequate capacity is available in the network. It will also investigate the possibility of connecting current residents on non-mains drainage	FRA Section 5

Figure i: Scoping Note Objective Table

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1 Introduction

1.1 Brookbanks Consulting Limited has been appointed by the Swindon EDA Consortium to complete a Flood Risk Assessment (FRA) for the proposed residential development consisting Phase I of the Swindon EV element of the wider EDA

1.2 Prior to undertaking this FRA, a scoping document was prepared and submitted to a working group of regulatory bodies, the aim being to agree the scope and fundamental outputs for this report and associated proposals. This document has since been approved by the Environment Agency and used to inform this FRA. The scoping note can be found in Appendix 2, along with the EA correspondence

1.3 The objective of the study is to demonstrate that the development proposals are acceptable from a flooding risk and drainage viewpoint. The proposals have therefore been developed having regard to the broad spectrum of published guidance such as (but not exclusively) NPPF, CIRIA 522, 609, 697 together with regional and local guidance in the form of the Swindon Borough Council Strategic Flood Risk Assessment et al.

1.4 This report summarises the findings of the study and specifically addresses the following issues in the context of the current legislative regime:

- flooding risk;
- surface water drainage; and
- foul water drainage.

1.5 Plans showing the existing and proposed development are contained in Appendix 1.

2 Background Information

Location & Details

2.1 The proposed development lies to the east of Swindon. The A420 and the Great Western Rail Line bound the site to the south. To the west of the site, Thornhill Lane forms the site boundary, with the extents of South Marston, Nightingale Lane and the accompanying farmland bounding the site to the north. Agricultural land forms the eastern boundary of the site.

2.2 The majority of the land is thought to be historically undeveloped, however, small developments including a combination of dwellings, hotels and public houses are spread around the site, with link roads included throughout. The site location and boundary is shown indicatively on Figure 2a, below.

Development Criteria

2.3 The following development profile is proposed within the 175 ha site.

- Up to 2,380 residential units
- Up to 2,500m^2 Retail
- Up to 2,500m^2 Commercial

2.4 Drawing 1335-SV-01 in Appendix 1 shows the overall development area

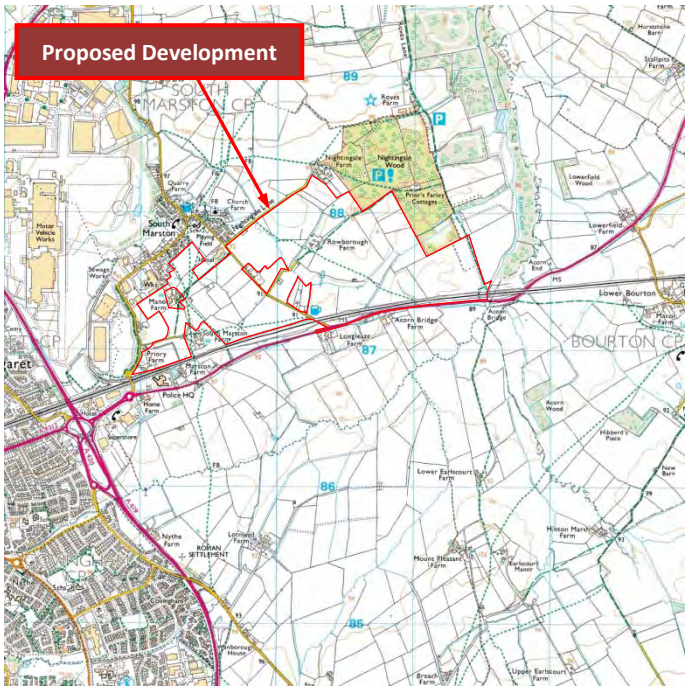


Figure 2a: Site location

Sources of Information

2.5 The following bodies have been consulted while completing the study:

- Thames Water - Storm & foul water drainage
- Environment Agency - Flood risk and drainage
- South Marston Parish Council - Flood risk and storm drainage
- Swindon Borough Council - Flood risk, drainage and associated policy

2.6 The following information has been available while completing the study:

- Site Survey - Brunel Surveys, 2007
- Flooding Information - South Marston Parish Council
- Published Geology - British Geological Society
- Upstream Pond Details - Honda UK
- Regional Flood Risk Assessment - South West Regional Assembly
- Stage 1 Strategic Flood Risk Assessment - Swindon Borough Council
- Water Cycle Study - Swindon Borough Council
- Preliminary Flood Risk Appraisal - Swindon Borough Council
- Draft Local Plan - Swindon Borough Council
- Infrastructure Delivery Plan - Swindon Borough Council

Topography & Site Survey

2.7 The South Marston Brook flows through the centre of the site forming a valley with topography characterised by relatively shallow and consistent falls, towards the brook from each of the site boundaries. Levels fall from a high point of 101m AoD on the north eastern corner of the site, adjacent to Nightingale Lane down to approximately 88m AoD on the eastern boundary adjacent to the South Marston Brook.

2.8 In the south eastern corner of the land, away from the South Marston Brook, a relatively small area of the site falls in an easterly direction towards the River Cole.

2.9 A copy of the site contours are contained in Appendix 1 as drawing 1335-SV-01.

Ground Conditions

2.10 Reference to the British Geological Society indicates a number of different geological strata across the site. To the south and west the site is reported to be predominantly underlain by Ampthill Clay formation mudstone with an outcrop of Ferruginous limestone on the western corner.

2.11 To the north and east, the BGS reports the general ground conditions to consist a mix of sandstone to the north east and limestone to the south.

2.12 With regards to superficial deposits, these are shown to be limited to a thin band of alluvium passing through the site, generally along the line of the South Marston Brook. The published geology is shown in Figure 2b, below.

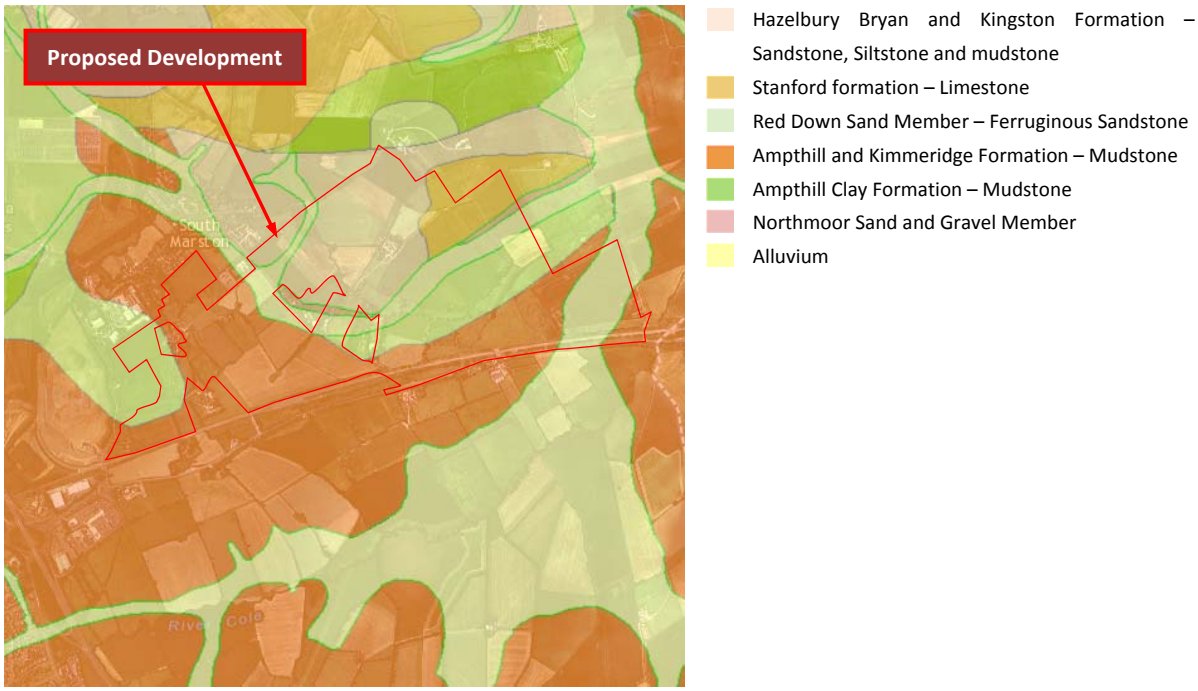


Figure 2b: BGS Published Geology

2.13 Site specific investigations across the wider site have yet to be completed however some localised borehole records have been obtained which generally confirm the ground conditions to be as per the BGS records.

Watercourse Systems & Drainage

2.14 The dominant watercourses in the area are the River Cole and South Marston Brook; the headwaters of the latter originate some 5km north west of the site in the Groundwell Industrial Estate.

Figure 2c: Watercourse Systems
River Cole
South Marston Brook

River Cole

2.15 Upstream of the site the River Cole flows through a large expanse of agricultural land south of the site generally in a north easterly direction after leaving the urban extent of Swindon. In this area the River Cole is relatively well defined and in the order of 3m – 5m wide for the most part.

2.16 Flowing in a north easterly direction from the A419, the River Cole has a confluence with a number of unnamed tributaries along with the Dorcan Stream near Longleaze Farm, all of which are in the order of 1.5m wide and appear more densely overgrown than the River Cole.

2.17 From here, the River flows in a northerly direction, passing beneath the railway line and A420 in a large pipe culvert. This is shown on Figure 2d below:

Figure 2d: River Cole Railway Culvert

2.18 Hereafter, the River Cole flows generally in a northerly direction circa 500m east of the eastern site boundary through mostly undeveloped land towards Sevenhampton, making confluence with the South Marston Brook near Lowerfield Wood.

South Marston Brook

2.19 The secondary watercourse in the area is the South Marston Brook. This watercourse originates some 5km north west of the proposed development site in the Groundwell Industrial Estate. From here, the brook flows generally in a south easterly direction through the residential area of Kingsdown before being culverted beneath the A419 and into the Honda Motor Vehicle Works.

2.20 The South Marston Brook emerges from this culvert on the eastern side of the Honda Motor Works and flows easterly towards South Marston. Prior to reaching the village, a pond with outflow control structure discharges to this watercourse as shown on figure 2e below.

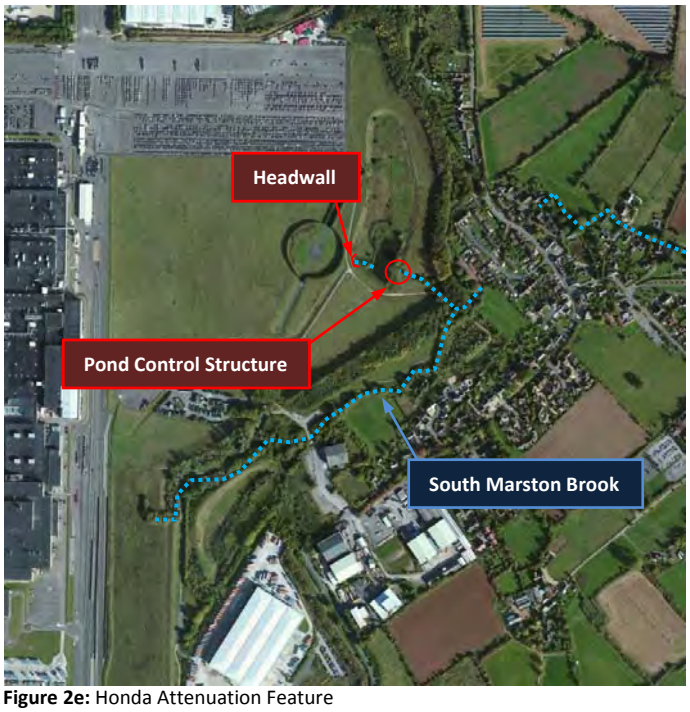


Figure 2e: Honda Attenuation Feature

2.21 Upon leaving the Honda area, the brook flows generally in an easterly direction through South Marston, culverted beneath the existing residential area and re-emerging as an open channel at Quarry Farm. At this point in its reach the south Marston Brook is in the order of 2.5m wide and is fairly well maintained, this is shown on Figure 2f below:

Figure 2f: South Marston Brook at Quarry farm

2.22 Hereafter, the brook turns by 90° and continues in a southerly direction, traversing the eastern extent of South Marston Village and thereafter flowing through the agricultural land comprising the proposed site before making confluence with the River Cole circa 2km east of South Marston Village near Lowerfield Farm.

2.23 Reference to the Flood Estimation Handbook CD dataset V3 shows the majority of the land to lie within the catchment of the South Marston Brook, passing through the east of the site. Having an URBEXT2000 value of 0.0669 the catchment can be described as *"slightly urbanised"*. The FEH catchment is shown in Figure 2g, below.

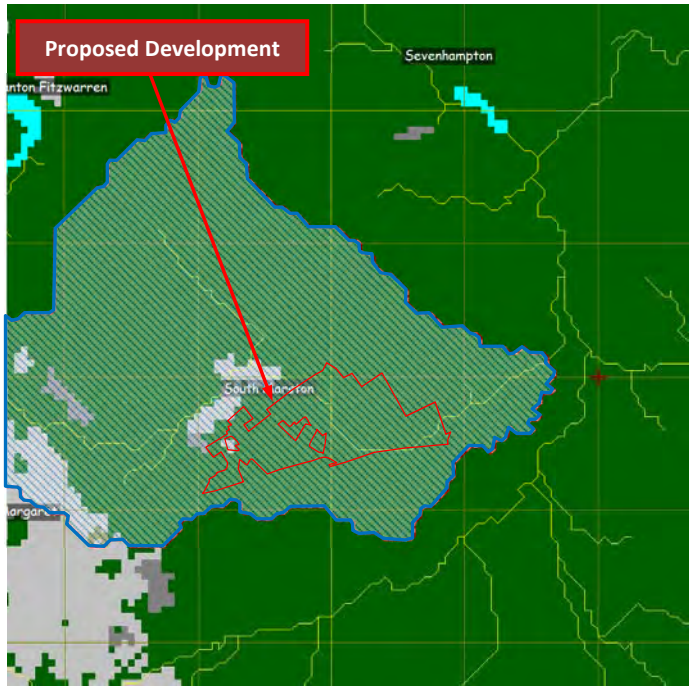


Figure 2g: FEH CD V3 Catchment

South Marston Flood Report

2.24 In July 2011, Simon Olive, working on behalf of South Marston Parish Council produced a detailed look at the areas surrounding South Marston, specifically concentrating on the land either side of Old Vicarage Lane and the impacts of surface water flows in this area.

2.25 The report, identifies key areas of concern, based upon local anecdotal evidence coupled with site inspection information.

2.26 Throughout this Flood Risk Assessment (FRA), this information has been used to help guide the assessments of flood risk at each stage with a summary of how this compares to published information is included along with how these sets of data have been incorporated within the site proposals.

3 Flooding Risk

National Planning Policy Context

3.1 The new National Planning Policy Framework (NPPF) was introduced in March 2012, which sets out Governmental Policy on a range of matters, including Development and Flood Risk. The policies were largely carried over from the former PPS25: Development & Flood Risk, albeit with certain simplification. The allocation of development sites and local planning authorities' development control decisions must be considered against a risk based search sequence, as provided by the document.

3.2 Allocation and planning of development must be considered against a risk based search sequence, as provided by the NPPF guidance. In terms of fluvial flooding, the guidance categorises flood zones in three principal levels of risk, as follows:

Flood Zone	Annual Probability of Flooding
Zone 1: Low probability	< 0.1 %
Zone 2: Medium probability	0.1 – 1.0 %
Zone 3a / 3b: High probability	> 1.0 %

Figure 3a: NPPF Flood Risk Parameters

3.3 The Guidance states that Planning Authorities should *"apply a sequential, risk-based approach to the location of development to avoid where possible flood risk to people and property and manage any residual risk, taking account of the impacts of climate change."*

3.4 According to the NPPF guidance, residential development at the proposed site, being designated as "More Vulnerable" classifications, should lie outside the envelope of the predicted 1 in 100 year (1%) flood, with preference given to sites lying outside the 1 in 1,000 (0.1%) year event and within Flood Zone 1

3.5 Sites with the potential to flood during a 1 in 100 (1%) year flood event (Flood Zone 3a) are not normally considered appropriate for proposed residential development unless on application of the "Sequential Test", the site is demonstrated to be the most appropriate for development and satisfactory flood mitigation can be provided. Additionally, proposed residential developments within Flood Zone 3a are required to pass the "Exception Test", the test being that:

- The development is to provide wider sustainability benefits
- The development will be safe, not increase flood risk and where possible reduce flood risk

Regional & Local Policy

3.6 **Regional Flood Risk Assessment:** In accordance with NPPF guidance, the South West Regional Assembly published their Regional Flood Risk Assessment (RFRA) February 2007. The document is a high level review of flood risk and strategy. In this document, concerns over the effects of potential of climate change are identified across Swindon and the wider south west region.

3.7 **Strategic Flood Risk Assessment:** To support local planning policy, NPPF guidance recommends that local planning authorities produce a Strategic Flood Risk Assessment (SFRA). The SFRA should be used to help define the Local Development Framework and associated policies; considering potential development zones in the context of the sequential test defined in the guidance.

3.8 Swindon Borough Council published their Level 1 Strategic Flood Risk Assessment in August 2008. The document generally underpins national guidance and provides recommendations to developers with regards to SuDS and design which will be explored further in this report under the **Storm Drainage** section

3.9 **Water Cycle Strategy:** In March 2007, Swindon Borough Council produced a Phase 1 Water Cycle Strategy (WCS). The objective of the strategy is to encourage an integrated and sustainable approach to the Water Cycle in Swindon and provide supporting evidence to inform the selection of preferred options for growth.

3.10 The study provided an overview of the foul drainage catchment for the wider Swindon area within the Thames Water catchment looking specifically at the capacity in both the network and treatment systems, along with water efficiency measures to reduce flows from developments and any upgrades required to accommodate for additional flows. This is further described in the F**oul Drainage** section.

3.11 Swindon Borough Council are currently in the process of producing a Level 2 Water Cycle Study which has further assessed the foul drainage situation in the area and it is envisaged that this will be published late 2013.

3.12 **Development Flood Risk Assessment:** At a local, site by site, level the NPPF guidance and supporting documents advocate the preparation of a Flood Risk Assessment (FRA). NPPF requires that developments covering an area of greater than one hectare prepare an FRA in accordance with Annex E of the guidance. The FRA is required to be proportionate to the risk and appropriate to the scale, nature and location of the development.

3.13 This document forms a Flood Risk Assessment (FRA), to accord with current guidance and addresses the requirements in demonstrating that the proposed development lies within the acceptable flood risk parameters.

Flood Mechanisms

3.14 Having completed a site hydrological desk study and walk over inspection, the possible flooding mechanisms at the site are identified as follows:

Mechanisms	Potential?	Comment
Fluvial (Annex C: C4)	Y	The River Cole and South Marston Brook lie within an influencing distance of the proposed development.
Coastal & tidal (Annex C: C5)	N	No tidal watercourses are close to the site
Overland flow (Annex C: C6)	Y	The published geology suggests the presence of underlying ground conditions which are unlikely suitable for infiltration and this the risk of overland flows must be considered.
Ground water (Annex C: C7)	Y	Geology underlying the site is of an impermeable nature, although topography does not necessarily suggest the likelihood of a problem. While not considered a high risk mechanism, ground water flooding needs to be considered.
Sewers (Annex C: C8)	Y	Whilst no sewers lie within the proposed development area, given the reported problems in the surrounding sewer network, the probability of a sewer flooding mechanism must be considered.
Reservoirs, Canals etc (Annex C: C9)	Y	A number of ponds have been identified which have the potential to impact upon the proposed development.

Figure 3b: Flooding mechanisms

3.15 Where potential risks are identified in Figure 3b, above, more detailed assessments have been completed and are outlined below. Further background is also outlined below.

Fluvial Flooding: C4

3.16 It is understood that the South Marston Brook was mapped as part of the National Flood Mapping Study completed by the Environment Agency. This study was based on the JFLOW model, the results of which are considered indicative only.

3.17 Therefore, whilst it is the responsibility of the Environment Agency formally to define the areas at risk of fluvial flooding from Main Rivers across the UK, as part of the ongoing site assessments, Bureau Veritas were appointed to complete a detailed hydraulic assessment of the River Cole and South Marston Brook.

3.18 A 2D hydraulic model of the South Marston Brook and the surrounding land was constructed using the TUFLOW hydraulic modelling package in order to calculate the displacement of water throughout a flood event by dividing the floodplain into a grid of individual cells. Each cell contains topographic information so flood flows follow any direction dictated by the local topography. The model therefore reflects morphological changes to the floodplain and the complex flowpaths present at the site.

3.19 The hydraulic model was constructed to conform to the Environment Agency's Strategic Flood Risk Mapping (SFRM) specification. Furthermore, the Environment Agency were consulted with regard to the development of the hydraulic model and their suggestions and comments incorporated into the model where appropriate.

3.20 The modelling exercise provided a full set of outputs for the following return events:

- 1 in 20 year
- 1 in 100 Year
- 1 in 100 Year +20%
- 1 in 1,000 Year

3.21 The full mapping completed following the modelling exercise is contained in Appendix 1 however an extract of the plan is shown as Figure 3c below:

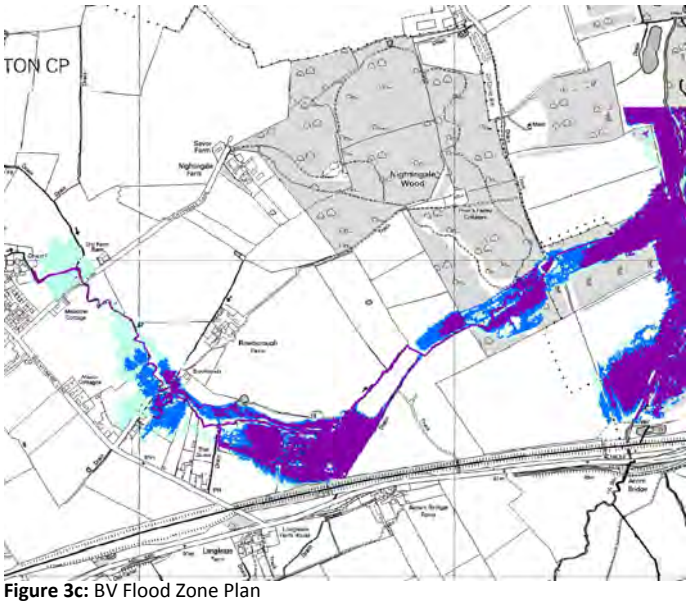
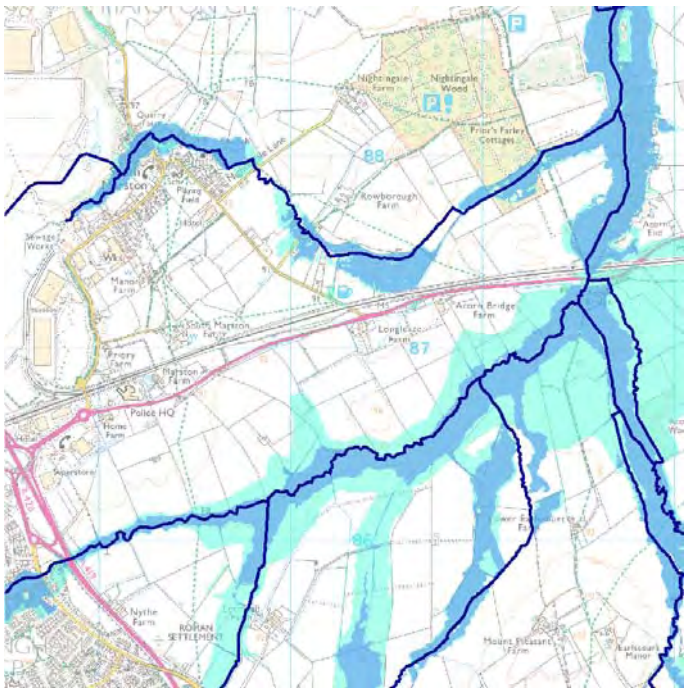


Figure 3c: BV Flood Zone Plan

3.22 The flood model identifies significant out of bank flows on the South Marston Brook passing through the site along with the River Cole to the east with a large area within the site boundary, adjacent to the rail line, shown to lie within Flood Zones 2 and 3.

3.23 Following completion of the modelling exercise, the flood model was submitted to the Environment Agency for review and approval. It was confirmed by the Environment agency on 11th October 2010 in their correspondence (ref: WA/2009/1016623/05-L01) contained in Appendix 1 that the model was acceptable and should be used for defining the flood risk parameters across the site.

3.24 Furthermore, the model was used by the Environment Agency to update their Flood Zone Mapping for the area, an extract of which is contained as Figure 3d below:



Flooding from rivers without defences – 1 in 100 year (1%) event (Zone 3)
Extent of extreme flood – 1 in 1,000 year (0.1%) event (Zone 2)
Flood defences
Areas benefiting from flood defences

Figure 3d: EA Flood Zone Plan

3.25 Photographic records of the flooding on the South Marston Brook have been provided by the Parish Council which generally correlate to the flooding identified through the modelling exercise.

3.26 The records are contained in Appendix 2 and have been used, where appropriate, further to define the flood risk parameters within the site boundary.

Coastal Flooding: C5

3.35 The site lies a significant distance from the nearest tidal watercourse and the coast. As such there is no risk of tidal or coastal flooding at this location.

Overland Flow: C6

3.36 Overland flow mechanisms result from the inability of unpaved ground to infiltrate rainfall or due to inadequacies of drainage collection systems to accommodate flow directed to gullies, drainage downpipes or similar. In minor cases, local ponding type flooding may occur. In more extreme events, flows accumulate and may be conveyed across land following the topography.

3.37 The Environment Agency publish a series of surface water flood maps for most areas across the UK which give an indication of the broad areas likely to be at risk of surface water flooding, i.e. areas where surface water would be expected to flow or pond.

3.38 In 2010 the Flood and Water Management Act defined 'surface runoff' as:

"Surface runoff" means rainwater (including snow and other precipitation) which—
(a)is on the surface of the ground (whether or not it is moving), and
(b)has not entered a watercourse, drainage system or public sewer."

3.39 Generally, the type of flooding shown by the Flood Map for Surface Water (FMfSW) produced by the Environment Agency, fits with the definition in the Act and shows the flooding that takes place from the 'surface runoff' generated by rainwater.

3.40 Two rainfall events, one with a 1 in 30 and the other with a 1 in 200 chance of occurring in any year, are modelled and mapped. For each rainfall probability, the mapping provides two layers of information which can be used individually to indicate:

- Surface Water Flooding' (flooding greater than 0.1m deep);
- Deeper Surface Water Flooding' (flooding greater than 0.3m deep).

3.41 The 0.3m threshold is chosen as it represents a typical value for the onset of significant property damages when property flooding may start (above doorstep level) and because it is at around this depth that moving through floodwater (driving or walking) may become more difficult; both of which may lead users to consider the need to close roads or evacuate areas.

3.42 Swindon Borough Council, as part of their duties as Lead Local Flood Authority, produced a Preliminary Flood Risk Assessment (PFRA) in August 2011.

3.43 The PFRA is a high level review that compiles information on significant local flood risk from past and future floods, based on current, readily available information.

3.44 The document specifically looks at local flood risk, being flooding originating from sources other than main rivers, the sea and large reservoirs and principally meaning flood risk from surface runoff, groundwater and ordinary watercourses however this does include:

- Lakes and ponds
- Flooding from sewers that is wholly or partly caused by rainwater or other precipitation
- The interaction with flooding from main rivers, the sea and sewers.

3.45 The Environment Agency has produced a national map of areas susceptible to surface water flooding for a 1 in 200 year (0.5% AEP) rainfall event which this has been utilised within the PFRA for Swindon as a baseline for the assessment of pluvial flooding in the borough.

3.46 Augmenting the EA mapping with local knowledge and additional sources of flooding information, the PFPA report culminated in a series of surface water flooding plans highlighting both the extent of surface water flooding and the likely depth for a 1 in 200 year (0.5% AEP) event. An extract from the plan is shown as Figure 3e below whilst the full plan is contained in Appendix 1.

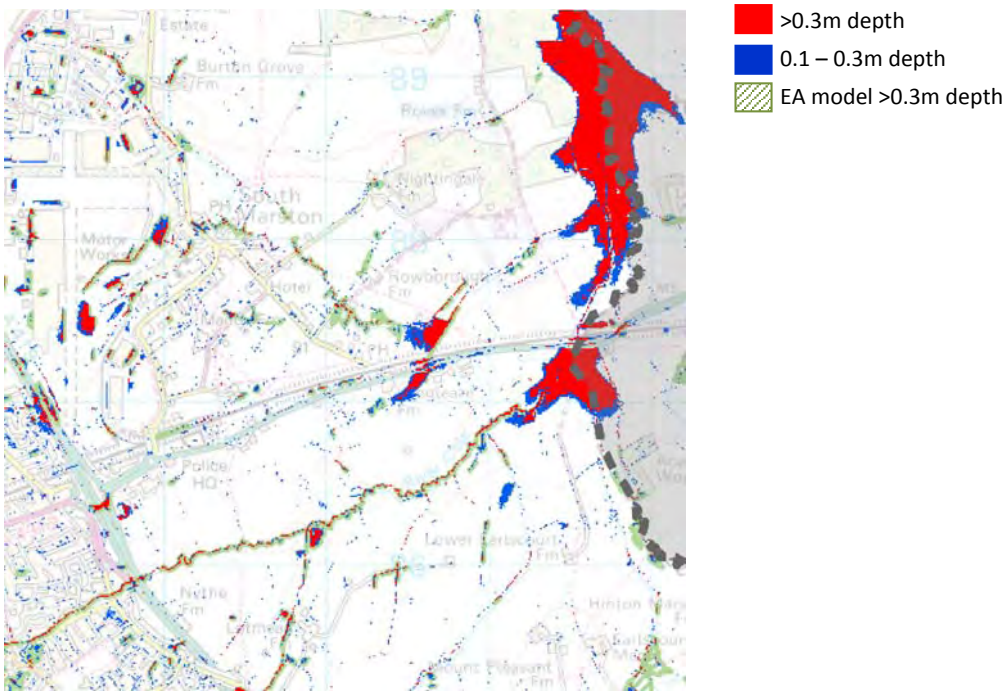


Figure 3e: Swindon PFRA Pluvial Flood Extents

3.47 The combined mapping produced by the Local Authority identifies a number of localised incidents of surface water flooding within the site boundary up to 0.3m in depth. Whilst much of the identified flooding is sporadic and minor in size, a large area of surface water ponding is identified on the banks of the South Marston Brook at Rowborough Farm.

3.48 Following discussions with the Environment Agency, Swindon Borough Council and South Marston Parish Council, further incidents of localised surface water flooding have been identified that are not identified on the fluvial flood plans generated by the EA, Borough Council or BV.

3.49 Anecdotal evidence in the way of photographic and video records have been provided to BCL which outline an area south west of the South Marston Brook to be liable to surface water flooding, this is shown illustratively on Figure 3f below whilst the photographs and their locations are contained in the Appendix 2.

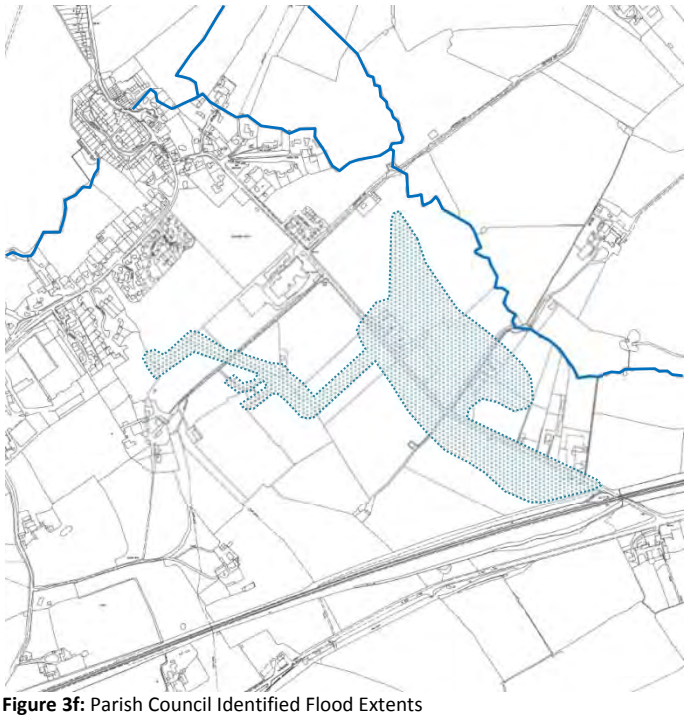


Figure 3f: Parish Council Identified Flood Extents

South Marston Brook & Tributaries

Parish Council Flood Extents

3.50 Whilst the information received by the Parish Council shows broadly the same areas as the EA mapping to be at risk, it has identified a more hydraulically connected area of surface water flooding across the agricultural land south of the village, flowing eastwards towards the South Marston Brook.

3.51 Further inspections show the outline provided above broadly correlate with the line of a number of shallow and poorly defined ditches bisecting these fields, this is shown on Figure 3g below:

Figure 3g: Existing ditch lines

3.52 Evidence from the site topographical data shows that it is likely that rainwater enters these ditches which flow towards the South Marston Brook however at key locations such as Old Vicarage Lane, these ditches become truncated.

3.53 As a result, the stormwater is held until the volume of water exceeds the storage capacity of the channels thus causing a surface water flooding mechanism as stormwater flows overland to reach the South Marston Brook following the topography.

3.54 Figure 3h below identified this flooding, clearly showing a linear area of flooding in the fields to the west of Old Vicarage Lane with the carriageway and residential properties forming a barrier to the conveyance of flow.

Figure 3h: Flooding at Old Vicarage Lane

Modelling Assessment

3.55 Whilst the EA and LPA have produced a number of risk based maps for the wider Swindon area, including the proposed development site, it has been requested that further modelling be undertaken using more detailed terrain mapping than the LiDAR data presently underpinning the mapping in order to quantify the risk posed by surface water run-off from the surrounding land.

3.56 A detailed two-dimensional overland flow model has therefore been created using the Infoworks ICM software. The key inputs for this are:

- Topography
- Rainfall profile

Topography

3.57 Detailed level information has been obtained for the wider South Marston and Rowborough areas in the form of Digital Terrain Mapping (DTM) data. This information has thereafter been used to define the prevailing ground levels in the area along with slope direction and gradient.

Rainfall Profile

3.58 The FEH CD contains digital catchment descriptors for over four million UK catchments, which drain an area of 0.5km^2 or greater. Additionally, the CD also provides rainfall depth-duration frequency information across the UK. The South Marston Brook is included in the FEH CD V3.

Assessment

3.59 To assess the risk to the proposed development from surface water run-off emanating from the off-site catchments, it is necessary to define the depth and velocity of potential overland flows during the critical 6 hour, 1 in 30 and 1 in 200 year storm events.

3.60 The rainfall profiles for these storm events have been derived from the FEH information through the Infoworks software and applied directly to the topographical data.

3.61 The assessments show a number of areas at risk of surface water flooding during both storm events which generally follow the routes of the River Cole and South Marston Brook with some minimal flooding identified broadly in those areas highlighted by the Parish Council. The surface water flooding mapping is contained in Appendix 1.

3.62 Following on from this, an assessment of hazard has been undertaken in accordance with EA Technical Document RnD FD2320/TR2 which confirms those areas of flooding, outside of the watercourse reaches, to be of a very low – low hazard. The hazard mapping is contained in Appendix 1 whilst an extract is shown as Figure 3i below.

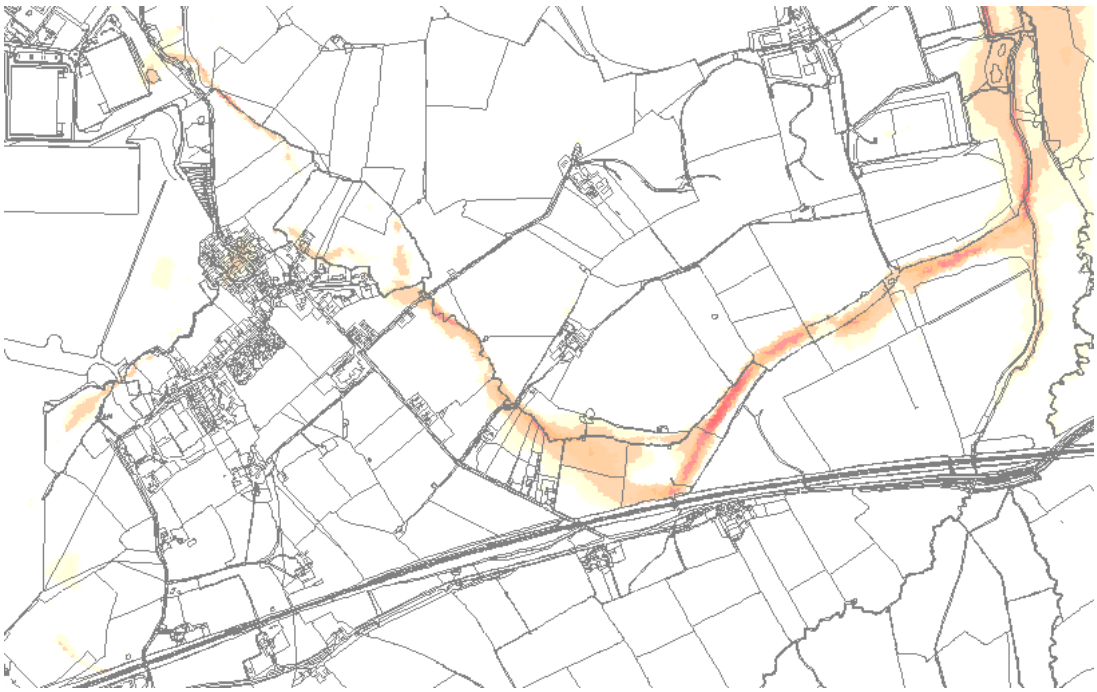


Figure 3i: Surface Water Hazard Mapping

Flood Hazard Rating	Colour Code	Hazard to People Classification
Less than 0.75		Very Low Hazard
0.75 to 1.25		Danger for Some – includes children, the elderly and the infirm
1.25 to 2.0		Danger for Most – includes the general public
More than 2.0		Danger for All – includes the emergency services

3.63 Whilst the above assessment confirms the risk posed by surface water run-off from outside the site to be **Very Low - Low**, by utilising both the modelling results and information obtained from the Parish Council, it will be possible further to provide positive environmental effects to the existing area by incorporating measures to reduce surface water flooding at this location.

3.64 The photos provided by the Parish Council are contained in Appendix 2 and are accompanied by measures to potentially alleviate each incident. It has been agreed by the Environment Agency and Swindon Borough Council that where these incidents lie outside of land in control, the development will not specifically provide the mitigation.

3.65 Furthermore, the introduction of positive drainage measures at the proposed site, including a number of interceptor and conveyance channels, will result in a reduced risk of surface water flooding from the proposed development to the existing and proposed properties by collecting and conveying stormwater within managed corridors and away from people and property.

Ground Water: C7

3.66 Ground water related flooding is fortunately quite rare, although the results of same, where present, can provide persistent problems that are problematic to resolve. Such mechanisms often develop due to construction activities that may have an unforeseen affect on the local geology or hydrogeology.

3.67 The Environment Agency mapping shows the sited to be underlain by both a minor aquifer of intermediate leaching potential in the east and west with a non aquifer formation found in the centre of the site.

3.68 Non Aquifers are formations of negligible permeability, usually regarded to contain insufficient quantities of groundwater. However, groundwater flow through non aquifers is a possibility even if a low one.

3.69 Minor Aquifers are fractured or potentially fractured rocks, which do not have a high primary permeability, or other formations of variable permeability including unconsolidated deposits. Although not producing large quantities of water for abstraction, they are important for local supplies and in supplying base flow to rivers

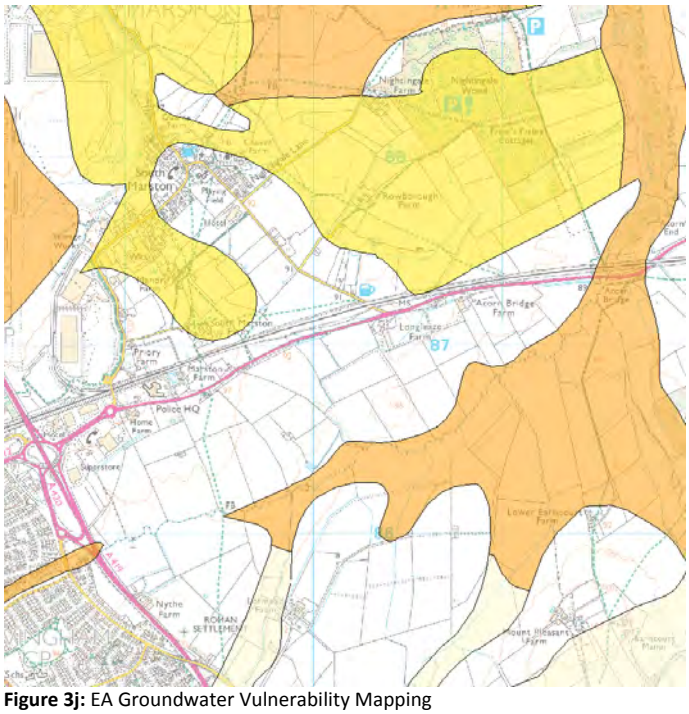


Figure 3j: EA Groundwater Vulnerability Mapping

3.70 Reference to the information provided as part of the Swindon Borough Council Level 1 SFRA does not identify any groundwater related flooding within the site boundary; indeed the nearest incident of groundwater flooding is circa 2km south of the site.

3.71 This incident occurred in Osprey Close on 18th December 2006 with no further groundwater flooding identified at this location since this time.

3.72 Further supporting this, borehole records in the area confirm the underlying groundwater to be in the order of 3.0mBGL. Groundwater flooding at this location is therefore considered to be of a low probability

Sewerage Systems: C8

3.73 Existing development to the north and east has a managed drainage system by way of sewers adopted by Thames Water and the Highway Authority.

3.74 The Swindon Borough Council SFRA provides evidence on local sewer flooding, which is taken from the Thames Water DG5 Asset Registers. Searches identify than no sewer flooding is identified within the proposed development site.

3.75 Historically, the area of South Marston to the north has suffered four incidents of sewer flooding as identified on the local authority SFRA mapping.

3.76 The SFRA further identifies that these flooding related incidents are a result of the inability of the existing foul drainage network in the area to deal with the volume of flows being received.

3.77 As part of the site development process Thames Water have been commissioned to complete a full modelling study of the existing foul network to identify where specific interventions may be required to not only facilitate the proposed development but to ensure no adverse impact on the surrounding area.

3.78 Furthermore, positive drainage measures incorporated on site, coupled with sustainable drainage systems (SuDS) will ensure that no increase in surface water will result from the site. Flood risk associated with sewer flooding is therefore considered to be a low probability

Artificial Sources: C9

3.79 Whilst no potential artificial sources of flooding have been identified within the site boundary, the Parish Council have identified a potential risk to the wider area from the Honda attenuation pond and as such, an assessment of same has been completed.

3.80 In 1990, Honda UK Manufacturing Ltd proposed the development of an assembly plant and associated infrastructure on the site of the former airfield to the north east of Swindon. As part of these proposals, Hydraulics Research were commissioned to complete a drainage assessment for the site to demonstrate that the new facility would not increase flood risk to the wider South Marston catchment.

3.81 It was proposed to implement an attenuation pond on the eastern boundary of the works, adjacent to the South Marston Brook which stored the run-off from the site during a range of return periods including the 1 in 100 year event to no greater than the current baseline.

3.82 The following design criteria for this attenuation pond was modelled:

Contributing Area	**-**	**0.73km^2**
100 Year Peak Flow	**-**	**5.52m^3/s**

3.83 The results of the modelling indicated that based on the following profile, the Honda pond will be able to reduce flows in the South Marston Brook to circa 4.71m^3/s, being a 15% reduction on existing flows.

Orifice outflow - 550mm Ø

Pond Level (mAOD)	Area (ha)
93.80	0.49
94.72	2.05
95.64	2.42
96.55	2.80

3.84 This drainage system was subsequently approved through the planning permission for the Honda UK works and it is believed to have been constructed in accordance with the Hydraulics Research documentation.

3.85 Further to assess the potential risk proposed by this attenuation pond during present day conditions, assessments have thereafter been completed using best practice methods by performing detention routing calculations for both the 1 in 1 and 1 in 100 year inlet and outlet return periods using the WinDES Source Control module. The summary calculations are contained in Appendix 2.

3.86 The calculations confirm that the existing pond performs adequately, retaining the 1 in 100 year flood event whilst allowing for approximately 1,000mm of freeboard between the peak water level and the top of bank. In order to provide a robust assessment, the following sensitivity tests were also completed:

- 30% allowance for climate change
- Potential growth of facility to 0.89km^2 (as outlined within original report)
- Growth with 30% climate change

Climate Change

3.87 By allowing for a 30% increase in flows due to climate change, in accordance with current best practice guidance, the pond performs satisfactorily, retaining stormwater without overflow whilst maintaining circa 600mm freeboard.

Facility Growth

3.88 With an increased contributing impermeable area of 0.89km^2, calculations demonstrate that water levels within the attenuation pond peak at 95.8mAOD, being 700mm below the top of bank level.

Facility Growth with Climate Change

3.89 This potential future growth, coupled with the effects of climate change have been modelled and confirm that the attenuation pond retains all floodwater without overflow and provides circa 250mm freeboard between peak water levels and top of bank.

3.90 The risk of flooding associated with artificial sources, including the Honda Pond is therefore considered low.

Parish Council Flooding Identification

3.91 As mentioned previously, South Marston Parish Council have provided a large amount of information relating to localised flooding incidents in the wider area, including within the proposed development site.

3.92 The photos provided by the Parish Council are contained in Appendix 2 and are accompanied by measures to potentially alleviate each incident. It has been agreed by the Environment Agency and Swindon Borough Council that where these incidents lie outside of land in control, the development will not specifically provide the mitigation. Figure 3k below provides a summary of these.

Photo Reference	Location	Description	Mitigation	Resolved by development
1	St Julan's	Manhole Blockage	Remove blockage	✘
2 - 3	Thornhill Rd Layby	Manhole Blockage	Remove blockage	✘
4 - 6	Hodinot Corner	Poor Connectivity	Improve connection to ditch	✘
7	Manor Park	Poor Connectivity	Improve connection to ditch	✓
8	Manor Park Pond	Missing Outfall	Confirm and upgrade outfall	✓
9 - 10	Manor Farm	Poor connectivity	Improve connection beneath road	✓
11 - 12	Mercure Hotel	Localised low spot	Improve drainage connectivity	✓
13	Playing Fields	Localised low spot	Minor ground reprofiling	✘
14	Playing Fields	Poor Connectivity	Improve connection to ditch	✘
15	Tennis Courts	No drainage	Provide drainage connection	✘
16	School Car Park	Manhole Blockage	Remove blockage	✘
17 - 18	Highworth Road	Poor Connectivity	Improve connection to river	✘
19 - 20	Highworth Road	Culvert blocked	Improve culverted sections	✘
21	Highworth Gardens	Poor Connectivity	Improve connection to ditch	✘
22 – 23	Old Vicarage Lane	Truncated ditch	Improve downstream connectivity	✓
24 – 25	Rowborough Lane	Poor Connectivity	Improve connection to ditch	✓
26 - 27	Rowborough Lane	Culvert blocked	Improve culverted sections	✓
28	Carpenters Arms	Culvert damage	Upgrade culverted sections	✓
29	Railway Fields	Poor Connectivity	Improve connection to ditch	✓
30	Sunnyside House	Fluvial Flooding	N/A	✘
31	Sunnyside Culvert	Fluvial Flooding	N/A	✘
32 - 33	Nightingale Lane	Fluvial Flooding	N/A	✘
34	Chapel Lane	Fluvial Flooding	N/A	✘

Figure 3k: Parish Council Flooding Locations

Flood Zoning & Probability Summary

3.93 In terms of fluvial flood risk, the majority of the proposed development lies in Flood Zone 1, being land that lies outside the 1 in 1,000 year (0.1%AEP) flood risk area. Those areas within the site lying within Flood Zones 2 and 3 will remain undeveloped with the exception of providing essential infrastructure routes through the development.

3.94 Assessment of other potential flooding mechanisms shows the land essentially to have a low probability of flooding from overland flow, ground water and sewer flooding.

3.95 Flooding from surface water mechanisms have been assessed as a low probability within the proposed development site with the modelling completed finding a small area within the proposed development to be at risk of same.

3.96 The risks of overland flow will be mitigated by providing adequate drainage systems in accordance with this document and utilising best practice measures as defined by local and national guidance.

Sequential Test

3.97 The NPPF Guidance states that Planning Authorities should complete a risk based "Sequential Test" at all stages of the planning process, to ascertain areas most suitable, from a flood risk perspective, for future development.

3.98 According to the planning policy statement, built development of a residential or educational nature, being designated as a "More Vulnerable" classification, should lie outside the envelope of the predicted 1 in 100 year (1% AEP) flood, with preference given to sites lying outside the 1 in 1,000 year (0.1% AEP) event and within Flood Zone 1.

3.99 Locations with the potential to flood during a 1 in 100 year (1% AEP) flood event (Flood Zone 3a) are not normally considered appropriate for proposed "More Vulnerable" uses unless on application of the "Sequential Test", the site is demonstrated to be the most appropriate for development and satisfactory flood mitigation can be provided.

3.100 While it is for Swindon Borough Council formally to apply the Sequential Test, the following paragraphs provide justification as to the appropriateness of development at Swindon EV in the context of the requirements of NPPF.

3.101 The essence of the "Sequential Test" is to "steer new developments to areas with the lowest probability of flooding" (Paragraph 101 of the NPPF guidance). The nature of the test will vary according to the status of the proposed development within the development plan system. Hence, "*for individual developments on sites allocated in development plans through the application of the Sequential Test, informed by a SFRA, developers need not apply the sequential test*" (NPPF guidance, Paragraph 104).

3.102 Of most relevance is Local Plan Policy EN6, which states:

"Policy EN6: Flood Risk

a. The risk and impact of flooding will be minimised through:

- *directing new development to areas with the lowest probability of flooding;*
- *ensuring that all new development addresses the effective management of flood risk;*
- *ensuring that development does not increase the risk of flooding elsewhere including on adjoining and surrounding land; and*
- *ensuring wider environmental benefits of development in relation to flood risk*

b. The suitability of development proposed in flood zones will be assessed using the Sequential Test, and, where necessary, the Exceptions Test. A sequential approach should be used at site level. *[our highlight]*

c. A site specific flood risk assessment will be required for development proposals of one hectare or greater in Flood Zone 1 and for all proposals for new development (including minor development and change of use) in Flood Zones 2 and 3 and Critical Drainage Areas, and also where proposed development or a change of use to a more vulnerable class may be subject to other sources of flooding. Appropriate mitigation and management measures must be implemented.

d. All development proposals must be assessed against the Swindon Surface Water Management Plan and the Local Flood Risk Management Strategy to address locally significant flooding. Appropriate mitigation and management measures must be implemented.

e. All development shall be required to provide a drainage strategy. Developments will be expected to incorporate sustainable drainage systems and ensure that run-off rates are attenuated to greenfield run-off rates. Higher rates would need to be justified and the risks quantified."

3.103 The principle of development at Swindon EV is already recognised in the Draft Local Plan, Policy SD2 states:

"**Policy SD2: The Sustainable Development Strategy**

a. Recognising its role and function in the wider area development in the Borough will be concentrated at Swindon through a combination of:

- *Realising development opportunities within Swindon's urban area; and*
- *Allocated strategic sites at; Wichelstowe, Commonhead, Tadpole Farm, Kingsdown and the proposed new Eastern Villages, Rowborough and expanded South Marston, as defined in the key diagram."*

3.104 As such in accordance with Policy EN6, the Sequential Test has already been completed satisfactorily by the local planning authority. In accordance with the policy, it is therefore not necessary for this FRA to provide a further Sequential Test assessment.

3.105 Notwithstanding this, the following paragraphs consider further evidence which provides a further endorsement as to the robustness of the proposed development in the light of the sequential approach advocated in NPPF Guidance.

3.106 The Swindon Borough Council published Draft Local Plan (December 2010) sets out a spatial vision for the Borough to aid decision making, The Spatial Vision states:

"The Borough of Swindon will become a place where people choose to live, visit and invest. As an important regional centre, Swindon's appeal will stem from having an attractive and well-equipped town that has successfully blended traditional architecture with high quality contemporary buildings that incorporate sustainable design and construction principles.

The achievement of a high quality public realm in the heart of the town linked to the countryside will provide a real focal point for visitors and Swindonians alike.

Swindon will become one of the best business locations in the UK, offering a high quality of life, not just to its residents, but also to those from a much wider catchment area. New jobs will be created and there will be the fullest range of employment opportunities for the whole community.

Swindon will be at the centre of a network of multifunctional green spaces linking the town to the wider countryside. Swindon will have responded to the needs of a growing population in a way that has protected and enhanced our natural and historic environment.

People in Swindon will have the opportunity to live active, healthy and learning lifestyles."

3.107 Furthermore, the draft Local Plan summarises ***"The Eastern Villages, Rowborough and South Marston are a key component of the development strategy for the Borough and is linked to many policies and strategies listed in the evidence base."***

Exception Test

3.108 While the Sequential Test should be employed on all sites to direct development to locations having the lowest probability of flooding, the Exception Test is to be used on select sites where it is shown necessary, through the Sequential Test, to apply development of a given Flood Risk Use Vulnerability in a Flood Zone that is ordinarily reserved for a less vulnerable use. Table 3 of the NPPF Technical Guide identifies four circumstances where the combination of a given Vulnerability of Use and Flood Zone justify the application of the Exception Test. Table 3 is reprinted below:

Flood Risk Vulnerability classification (see Table D2)		Essential Infrastructure	Water compatible	Highly Vulnerable	More Vulnerable	Less Vulnerable
Flood Zone (see Table D.1)	Zone 1	✔	✔	✔	✔	✔
	Zone 2	✔	✔	Exception Test required	✔	✔
	Zone 3a	Exception Test required	✔	✗	Exception Test required	✔
	Zone 3b 'Functional Floodplain'	Exception Test required	✔	✗	✗	✗

Figure 3e: NPPF Technical Guide Table 3

3.109 With a small area of land within the site lying in Flood Zone 3, the development is required to pass the Exception Tests for one Flood Risk / Vulnerability Classification; being:

- Highways within the site being classified as 'Essential Infrastructure' in Zone 3a

3.110 To pass the Exception Test, Paragraph 1002 of he NPPF guidance requires that:

a) ***It must be demonstrated that the development provides wider sustainability benefits to the community that outweigh flood risk, informed by a SFRA where one has been prepared. If the DPD has reached the 'submission' stage – see Figure 4 of PPS12: Local Development Frameworks – the benefits of the development should contribute to the Core Strategy's Sustainability Appraisal***

As with the Sequential Test, Paragraph 20 of PPS25 urges that the Exception Test be applied early in the process where this is possible.

In this regard it is highly relevant that the flood risks apparent within the Swindon EV were very much taken into account in the formulation of the Draft Local Plan and in the identification of site as a key component of the wider development strategy for Swindon. Paragraph 3.106 of this FRA draws attention to the conclusions of the Draft Local Plan with regard to the Swindon EV development. Without re-quoting the findings of the document, in summary, the proposed Eastern Villages (the application site) was considered a highly suitable future opportunity to provide much needed housing opportunities.

With the Councils conclusions, supported by a Flood Risk and Sustainability Appraisal, it may be concluded that the development of an urban extension at Swindon EV fulfils the requirements of the Exception Test having been demonstrated to deliver wider sustainability benefits that outweigh a flood risk that was taken fully into account at that strategic scale.

b) ***A FRA must demonstrate that the development will be safe, without increasing flood risk elsewhere, and, where possible, will reduce flood risk overall.***

This FRA considers flood risk to both proposed development and the wider South Marston and Rowborough catchment in a comprehensive manner. Multi mechanism flooding is considered and it is demonstrated the proposed development will be safe.

On and off-site flood mitigation improvements identified on the flooding photo matrix will ensure the development helps provide a reduction in flood risk across the wider Eastern Villages area by obviating, where possible, the effects of existing flooding mechanisms. Residual Flood Risk relating to the proposed development is outlined below.

Residual Flood Risk

3.111 This FRA outlines various protection measures to manage residual flood risk. For example freeboard proposals are outlined that provide a factor of safety to raise floor slab levels of buildings above the existing ground levels and above predicted fluvial flood levels when climate change is included. Proposals are also outlined to ensure the development complies with current guidance on flood resilient construction by providing infrastructure routes through the development that would channel all but extreme flood waters in a controlled manner through the development and away from property.

3.112 In summary, the following residual flood risk controls are also proposed:

- **Information Pack:** Production of a welcome pack / introductory brochure for all residents in order to outline the procedures for dealing with flooding and what to do in the event of a flood.

- **Maintenance:** Provision of a 9m maintenance strip to all watercourses presently maintained by the EA along with provision of an agreed maintenance strip to all new features such that adequate maintenance can be achieved.

- **Maintenance:** Provision of ongoing maintenance for SuDS features, ordinary watercourse and existing artificial water bodies.

- **Freeboard:** the site will deliver 150mm freeboard above the above the nearest bank level of the South Marston Brook on-site.

- **Routing:** Compliance with guidance in terms of flood routing and resilience for new developments, employing infrastructure corridors safely to covey any extreme event flows / blockage away from properties.

- **SuDS:** Provision of a storm water SuDS management system (See Section 4).

- **Drainage:** Connection to a point of adequacy on the foul water drainage network with completion of necessary downstream reinforcements to ensure adequate conveyance and treatment capacity. (See Section 5).

- **Drainage:** Adoption and associated ongoing maintenance of development storm and foul drainage system.

3.113 The proposed residual flood risk management measures will provide safe homes and facilities at Swindon EV.

Summary

3.114 All built development, with the exception of essential infrastructure, lies in fluvial Flood Zone 1 and hence has a low probability of flood risk.

3.115 Assessment of, ground water and sewer flooding mechanisms does not suggest other than a low risk of flooding at the site.

3.116 Assessment of the overland flow mechanism finds the site to be at very low risk, furthermore and potential mechanism will be adequately mitigated post development.

3.117 It may be concluded that the proposed development complies NPPF guidance in terms of appropriateness of the land use and flood zoning.

Objectives

3.118 The key development objectives that need to be achieved in relation to flooding are essentially listed above in the sub-section entitled Residual Flood Risk.

4 Storm Drainage

Background

4.1 To understand the baseline provision for storm drainage in the area, a copy of the Thames Water sewerage network records has been obtained. Public storm water sewers are present within the South Marston residential area to the north west of the proposed development.

4.2 The land is presently not serviced by a positive storm water drainage network. It is believed that storm water currently discharges to the South Marston Brook passing through the site.

Drainage Options

4.3 The following paragraphs in this section outline the proposed drainage strategy to meet national and local design requirements and guidance.

4.4 Current guidance[1] requires that new developments implement means of storm water control, known as SuDS (Sustainable Drainage Systems), to maintain flow rates discharged to the surface water receptor at the pre-development 'baseline conditions' and improve the quality of water discharged from the land.

4.5 The Swindon Borough Council SFRA underpins national guidance on the provision of storm water drainage, encouraging the use of sustainable means of drainage at new developments.

4.6 As part of the detailed scoping exercise for the proposed development, it was agreed that the development would seek to reduce site discharge to below the baseline rate during the critical 1 in 100 year storm as calculated using the IoH124 methodology in accordance with national guidance, thus providing a significant attempt to reduce flood risk across the wider catchment.

4.7 It is proposed to implement a multi tiered SuDS scheme consistent with local and national policy at the proposed development. By implementing a range of SuDS features and utilizing best practice design and construction methods it will be possible to mitigate any overland flow flood risk within the site boundary.

4.8 When appraising suitable storm water discharge options for a development site, Part H of the Building Regulations 2002 (and associated guidance) provides the following search sequence for identification of the most appropriate drainage methodology.

"Rainwater from a system provided pursuant to sub-paragraphs (1) or (2) shall discharge to one of the following, listed in order of priority -

(a) an adequate soakaway or some other adequate infiltration system; or where that is not reasonably practicable,

(b) a watercourse; or where that is not reasonably practicable,

(c) a sewer. "

4.9 Dealing with the search order in sequence:

[1] NPPF, CIRIA C522, C609, C697 et al.

(a) Source control systems treat water close to the point of collection, in features such as soakaways, porous pavements, infiltration trenches and basins. The use of same can have the benefit of discharging surface water back to ground rather than just temporarily attenuating peak flows before discharging it to a receiving watercourse or sewer.

As source control measures commonly rely upon the infiltration of surface water to ground, it preferable that the ground conditions are appropriate for such. Site specific ground investigations have yet to be completed however previous experience of the underlying geology suggests the site may not be suitable for a wholly infiltration based drainage strategy.

Through scoping of this FRA, a commitment has been made to providing source control measures has been made, which will limit the reliance on larger secondary and tertiary treatment systems to achieve the appropriate design criteria. Measures such as permeable paving, filter strips and conveyance strips will be utilised in unlined form, where possible, to provide the opportunity of achieving some storm water infiltration at the site.

Detention and conveyance systems placed close to source may can become important source control features, aiding conveyance between the source control features and the localised detention system.

(b) Next in the search sequence, defined by Part H, is discharge to a watercourse or suitable receiving water body. Where coupled with appropriate upstream attenuation measures, this means of discharge can provide a sustainable drainage scheme that ensures that peak discharges and flood risk in the receiving water body are not increased.

The South Marston Brook passing through the site provides an appropriate receptor for storm water from the development. In accordance with national policy, proposals to discharge storm water from the proposed development to the watercourse should at least maintained the status quo in terms of peak discharges, but preferably provide some betterment by way of a reduction in peak flows discharged to the receptor.

(c) Last in the search sequence is discharge to a sewer. In the context of SuDS this is the least preferable scheme as it relies on 'engineered' methods to convey large volumes of water from development areas, has a higher likelihood of flooding due to blockage and provides less intrinsic treatment to the water.

Thames Water records show that no public surface water sewers are available within the site that could be employed by the development. Existing storm sewers are present to the north and west of the proposed development. As a SuDS strategy is proposed whereby discharge will be directed to the ordinary watercourses on-site, at this stage a connection to the Thames Water is not envisaged to be required.

4.10 The search sequence outlined above demonstrates that a site drainage system providing a storm water management train, involving source control and discharge to a watercourse is the most appropriate means to control storm water from the development.

4.11 Scoping discussions have also agreed that water management system should not rely on a small number of large detention basins lying at the low point of the catchment. In the alternative, the requirements of the regulatory authorities has been for a treatment system to be developed that provides source control, secondary and tertiary treatment systems, thereby avoiding reliance on a number of large features and limiting the reliance on very large piped drainage systems.

4.12 Accordingly, proposals have been developed to inform the strategic drainage network across the development. It is proposed that the drainage system for the site utilises a multi tier SuDS treatment train system.

4.13 Coupled with the storm water flow control benefits, the use of a multi tier SuDS treatment train based approach to drainage can also provide a betterment on water quality. National guidance in the form of CIRIA 609 outlines that by implementing a *"SuDS Treatment Train",* stormwater from the site can be polished to an improved standard thus ensuring the development proposals have no adverse affects on the wider hydrology.

4.14 The following paragraphs outline the potential SuDS features appropriate for use on site and their place within a multi tiered system.

Primary Drainage Systems (source control)

4.15 At the head of the drainage network, across the site, source control measures will be implemented to reduce the amount of run-off being conveyed directly to piped drainage systems. As site specific infiltration testing has shown the ground to be practical impervious, source control will be limited to detention type systems, albeit that systems will be unlined and therefore provide for an element of infiltration. The Primary Drainage System will reduce the initial peak discharge at a local level before discharge into the site Secondary Drainage System.

4.16 Through work on other similar strategically sized projects, BCL has shown that peak discharges of circa 15% in residential areas can readily be achieved without unacceptable reductions in net developable land or prohibitive financial implications. It is therefore proposed to set these values as a benchmark for each parcel of the development.

4.17 The primary aims of the Primary Drainage System will therefore be:

- Reduction in peak discharges by 15% in residential areas.
- Provide one stage of water quality treatment where appropriate

4.18 Through consultations at pre-application stage, it has been agreed that nature of source control measures to be implemented will need to remain flexible, providing each house builder with a 'toolkit' of options to reach an agreed target for peak discharge reduction and water treatment. The following paragraphs describe a number of options available.

Permeable Paving

4.19 Permeable Paving is approved by many Local Authorities for implementation on shared surfaces and in some cases, development road networks and can and can act as a receptor for surface water run-off from nearby house roofs. However, the system is perhaps best suited to managed parking areas and shared surfaces where block paving is typically used as the surface treatment and ongoing maintenance can be ensured by way of a management company or the like.

4.20 There is little need for underground pipes or gullies, and the attenuation afforded within the sub-base layer helps to reduce the volume of storage required elsewhere.

Figure 4a: Permeable paving

Filter Strips

4.21 Filter strips have been used in the drainage of highways alike for many years. The absence of traditional pipe work in such a system frees the drainage design to employ shallow gradients on both channels and drains, which in turn also act as a means of passive treatment to improve water quality.

4.22 Highways within the development could potentially be amended to incorporate filter drains. Alternatively, filter strips can be used to collect flows from areas such a group of house. Figure 4b below shows an example of a filter strip in a road corridor.

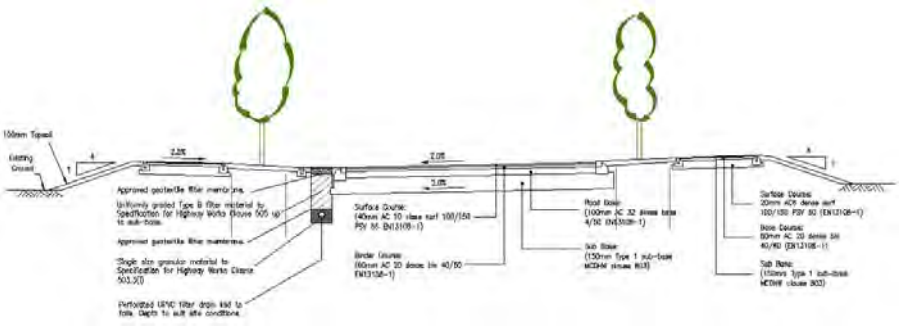


Figure 4b: Filter Strip along highway

Ditches

4.23 Ditches may be used along highways and in common areas to infiltrate, attenuate and convey flows from hard surfaces across the development before being discharged in to the secondary system. Linear features, such as ditches and filter strips provide an efficient means of improving water quality.

Swales

4.24 While swales implemented at development parcel level can be very land hungry, costly to maintain and provide difficulties with frontage access, the opportunity potentially exists to implement swale on a number of strategic highway routes, with no direct frontage, through the development. Green space being incorporated along the highways could be designed to allow 'over the edge' flows to be directed into the swale for infiltration, attenuation and conveyance before discharge to the secondary system. A typical highway swale is show in Figure 4d below:

Figure 4d: Swale along road corridor

Secondary Drainage System

4.25 The secondary drainage system will collect partially treated, excess, water from the primary, source control system at a local level, thereafter providing both flow and water quality attenuation and flow conveyance through the site towards the main outfall.

4.26 As a strategic element of the site storm water management system, servicing the site for development, the basic form of the secondary system has been designed and incorporated into the masterplan. The secondary system is made up of a series of rural nature channels traversing the boundary of the build development along the linear parkland.

4.27 The channels form an integral element of the masterplan, providing both function and form within the layout. The channels will be formed at very low gradients within the green areas in the heartland of the development, providing a series of controls and cascades for water to collect and flows covey down the system.

4.28 The primary aims of the channels are:

- Provide as much flow attenuation and water quality treatment as possible within the reasonable constraints of the layout
- Reduce piped flow lengths from the primary system
- Convey flows to the tertiary treatment system
- Provide a flow corridor for potential exceedance flows through the site.
- Provide landscaping and amenity benefits incorporated into the area such that finished arrangement will be pleasing and provide valuable communal areas across the development.

4.29 Figures 4d below provide an illustrative representation an attenuation channel:

Figure 4d: Development Edge Channel

Tertiary Drainage Systems

4.30 The tertiary drainage system will provide the final stage of treatment to storm water before flows are conveyed to the point of discharge from the site drainage network. A series of detention features will be provided with the primary functional aim of completing the final element of flow and water quality conditioning.

4.31 The proposed basins are to be located within or on the fringes of the proposed country park, and provide on-line storage, receiving surcharged flows from the secondary drainage system in high rainfall events. Being the last stage of treatment, only the more extreme events will result in the basins extending to temporarily store run-off from the development.

4.32 It is anticipated that basins will primarily be normally dry with permanently wet low flow channels to convey run-off in periods of low rainfall, which will in turn provide the passive treatment benefits offered within the remainder of the surface water management network.

4.33 The primary aims of the tertiary basins will therefore be:

- Final flow and water quality conditioning to meet the agreed design criteria
- Provide landscaping, amenity and ecological benefits within the parkland area of the site.

Figure 4e: Storage Basin

Preliminary Drainage Design

4.34 Preliminary engineering design the site drainage system, outlined above, has been completed to inform principles of the system for masterplanning and to demonstrate the viability of the storm water management train to achieve the design criteria.

4.35 Preliminary assessment of the requirements for storm drainage has been based on the following criteria:

Application Site Area:	175.0 ha
Developed Area:	74.59 ha
Landscaped Area:	100.41 ha
Impermeability – Residential[(2)]:	0.55
Impermeability - Primary school[(3)]:	0.45
Sewer design return period[(3)]	1 in 1 years
Sewer flood protection[(2)]	1 in 30 years
Fluvial / Development flood protection[(1)]	1 in 100 years
M5-60[(4)]	20.0 mm
Ratio r[(2)]	0.40
Minimum cover to sewers[(1)]	1.2 m
Minimum velocity[(1)]	1.0 m/sec
Pipe ks value[(1)]	0.6 mm
Allowance for climate change[(5)]	30%

Baseline Run-off

4.36 National policy[1] dictates that new developments control the peak discharge of storm water from a site to the baseline, undeveloped, site conditions. Over very large development areas, the baseline rate of run-off is normally estimated using the FEH methodologies. However, Paragraph 3.1.2 of the FEH guidance states:

"The frequency estimation procedures can be used on any catchment, gauged or ungauged, that drains an area of at least 0.5km^2. The flood estimation procedures can be applied on smaller catchments only where the catchment is gauged and offers simple flood peak or flood event data"

4.37 On undeveloped and ungauged catchments of less than 0.5km^2 in area, it is correct to complete baseline site discharge assessments using the nationally accepted IoH124 methodology for small rural catchments. Local policy is to employ IoH124 in a manner set out by the Environment Agency document R&D Technical Report W5-074/A/TR/1 Revision D: Preliminary Rainfall Runoff Management or Developments and CIRIA C697. This methodology requires that, for catchments of less than 50ha, the IoH assessment is completed for a 50ha area with the results linearly interpolated to determine the flow rate value based on the ratio of the development to 50ha.

4.38 Pre-application consultations with the with EA and Swindon Borough Council have identified an aspiration to provide a post development run-off from the site that is somewhat below the baseline, present day, rate; the intention being that positive environmental effects result by way of reduced flood risk across the catchment. Accordingly, it has been agreed, through formal scoping, that the storm water run-off from the proposed development will be reduced on the calculated IoH124 run-off from the site.

[2] Impermeability ratios are taken from historic measured figures on similar Brookbanks development types
[3] Sewers for Adoption 6th Edition
[4] Wallingford Report
[5] NPPF Guidance